Knowledge-Grounded Voice AI Platform

Written Project Walkthrough (submitted in place of a recorded video)

Repository: https://github.com/diwandahiya304/Darwix-AI-

1. System Overview

This is a knowledge-grounded voice AI platform covering all four parts of the assessment. Everything shares one retrieval API built in Question 2 (kb/). The voice agent in Question 1 (voice_agent/) grounds every answer against it. The two multilingual bots in Question 3 (multilingual/, Philippines + Indonesia) reuse the same index with language-tagged records. Question 4 (live_nudges/) is a separate real-time pipeline that turns live call audio into in-call nudges.

The single rule that runs through everything: the bot must say "I don't know" instead of inventing an answer -- checked by an automated test in every voice component, not just prompted for.

2. Architecture and Key Design Decisions

High-level flow:

- kb/ (Q2): ingest -> clean -> mask PII -> dedupe -> chunk -> version -> hybrid dense+BM25 index -> FastAPI /search
- voice_agent/ (Q1) and multilingual/ (Q3) both call kb/'s /search directly, filtered by language -- never hardcoded facts in a prompt
- live_nudges/ (Q4): independent pipeline -- real synthesized call audio -> real streaming ASR -> signal extraction -> nudge controller -> WebSocket dashboard / CLI

Three decisions mattered more than the rest:

Hybrid retrieval (dense + BM25), not dense-only.
Pure embedding search sometimes ranked a loose paraphrase above an exact keyword match (a policy number, a product name). BM25 catches what embeddings miss; fusing both with Reciprocal Rank Fusion covers more real phrasing than either alone.

Per-language grounding thresholds, not one global number.
The same relevance threshold that worked for English was measurably too strict for Filipino and Indonesian content -- calibrated separately for each, against real measured score distributions, not guessed.

A real regression found and fixed: merging Q3's multilingual KB records into the shared index silently broke Q1's English hallucination check (88% to 62% pass rate), because searches were not language-filtered. Fixed with an explicit language filter everywhere, locked in with a regression test.

3. Knowledge Base and Retrieval Design (Q2)

Schema matches the brief's field example exactly: record_id, title, content, category, source_ref, version, pii -- every record traceable to a source. Chunking is category-aware: FAQ/objection content splits into Q/A pairs, eligibility rules split by bullet, everything else splits by heading then sentence-count. PII is identified and masked with Presidio plus custom Indian-ID recognizers.

Real measured results (docs/q2_kb/retrieval_test_results.json):

- hit@1 = 1.00, hit@3 = 1.00, out-of-scope rejection = 1.00 (13 queries)
- Retrieval latency: P50 = 74.1ms, P95 = 103.4ms -- well under a 300ms target

- Working connection to Q1 confirmed: real HTTP calls, not hardcoded FAQs

4. Voice Agent Flow (Q1)

The system prompt contains zero facts -- verified by an automated test. Every fact comes from a search_knowledge_base tool call at answer time. Qualification scoring, escalation, and lead creation are deterministic code, not the LLM guessing -- important for auditability in an insurance context.

Real results:

- All 5 required scenarios pass: cooperative, objection, incomplete/conflicting details, out-of-scope, human escalation
- Hallucination refusal rate: 87.5% (7/8) -- one genuine, root-caused miss, reported honestly, not excluded

Honest gap: no live Vapi account exists in this environment, so there is no callable number or web-calling interface right now. These are simulated calls against real webhook/tool/KB code, not recorded phone calls.

5. Multilingual Handling (Q3)

Two markets reuse the same KB and retrieval code with language-tagged records and their own calibrated grounding thresholds.

- Philippines: life insurance/bancassurance, English/Tagalog/Taglish, register-mirrored (formal po/opo vs. casual). 100% grounding rate across 5 scenarios.
- Indonesia: multifinance, formal + colloquial Bahasa + English loanwords, OJK-style compliance guardrails in the system prompt. 86% grounding rate across 6 scenarios (one genuine paraphrase-distance retrieval miss, root-caused not hidden).
- At least 3 localization (not translation) examples per market are documented.

Honest gaps: the Indonesian regional accent is text-level only (no audio was synthesized for Q3 at all), and the ASR report for Q3 is explicitly illustrative-only -- no Deepgram/Azure credentials, no real audio, no real recorded calls for this market.

6. Live Nudges (Q4) -- the most real part of this submission

This is a genuinely real pipeline, not simulated. A local text-to-speech engine synthesizes 4 scripted scenarios into real stereo call audio (agent on the left channel, customer on the right). That audio is replayed in real, paced 300ms chunks -- never analyzing a finished recording. A real local speech-to-text model (faster-whisper) runs genuine inference per rolling window as the audio plays. Two signal layers (fast keyword rules, plus an optional LLM layer that honestly reports itself as "not configured" rather than faking a result) feed a nudge controller with per-type confidence floors, cooldowns, a semantic-similarity duplicate check, expiry, and priority ordering (compliance beats frustration beats opportunity). Delivery is a WebSocket dashboard and a CLI.

Real measured results:

- Nudge quality (docs/q4_live_nudges/quality_report.md): 83% precision, 100% recall, 0 false negatives, across 4 scenarios x 3 runs each. The one recurring false positive is explained, not hidden: a casual 'sounds good' collides with the buying-signal keyword list.
- Latency (docs/q4_live_nudges/latency_report.md): ASR P50 = 834ms, P95 = 1651ms (real local model inference); signal extraction and the LLM layer both under 2ms P95.
- A real (not illustrative) ASR word-error-rate was measured: 44.9% mean WER against synthetic TTS voice audio -- traced to two specific, documented causes (number-format mismatches and a stereo-channel-bleed hallucination artifact), not just reported as a bare number.

Building this surfaced and fixed several real bugs: a sentence split across a fixed ASR window boundary silently missed a compliance signal; trailing audio was being silently dropped when a call ended mid-window; and an early

latency-measurement bug made the real ASR inference time vanish from the report. All are logged with how they were found and fixed.

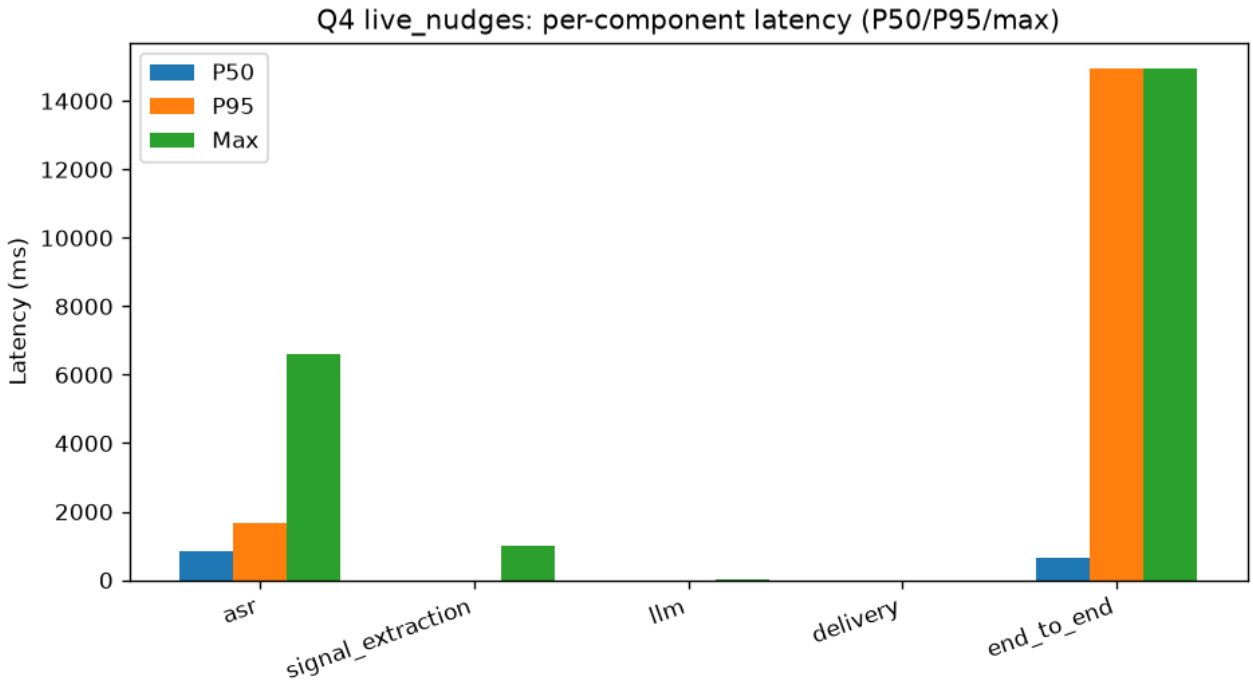

Figure: Q4 per-component latency (P50 / P95 / max), from docs/q4_live_nudges/latency_chart.png

7. Error and Fallback Cases (across all components)

- Q1: out-of-scope questions get a polite decline and redirect, never a half-answer (recordings/voice_agent/d_out_of_scope.md)
- Q3: below the per-language grounding threshold, the bot states it doesn't have that information, in the customer's own language and register -- never switching to English unexpectedly
- Q4: the noisy/ambiguous scenario produces zero or one nudge, not a flood of false positives; a low-confidence signal is genuinely suppressed by the controller and logged with its reason

8. Known Limitations

- No live Vapi/Deepgram/Azure/Anthropic credentials anywhere in this environment by default -- Q1 has no callable number, Q3 has no synthesized audio at all, Q4's LLM layer and telephony side are unconfigured
- Q4 does not reuse Q1/Q3 recordings (allowed, not required, by the brief) -- it has its own independently-synthesized audio
- Cross-lingual retrieval margins are thin -- a real regression (found and fixed) shows how easily this class of bug reappears
- Q4's ASR uses fixed-timer windows, not voice-activity detection, and is measurably imperfect (44.9% WER) even against robotic TTS audio
- Q4's single-process ASR model and in-process WebSocket broadcaster don't scale to 10x concurrent calls without a real queue/pub-sub backbone
- No native-speaker or compliance/legal review of any Filipino/Bahasa Indonesia content, or of the OJK/IRDAI-style guardrails

9. Production Improvement Plan

- Wire in real providers (Vapi, Deepgram/Azure, Azure/ElevenLabs TTS) behind the existing interfaces -- no code shape change needed, only credentials plus a live smoke test

- Monitoring and observability: structured logging for every KB search, every tool call, every nudge fired/suppressed, plus latency/error dashboards
- A continuous eval pipeline in CI (retrieval hit@k, hallucination refusal rate, nudge precision/recall) with a required-to-pass threshold, instead of on-demand reports
- PII compliance hardening: multilingual NER validated for Filipino/Bahasa Indonesia, encryption at rest, an audit log, a data-retention policy
- Cost tracking: LLM tokens, ASR minutes, TTS characters, with a per-call cost ceiling and alerting
- Scaling: per-call-id state and a real pub/sub backbone (Redis/Kafka) for Q4, horizontal scaling for the KB API
- A human-in-the-loop review queue for hallucination-check failures, suppressed nudges, and retrieval misses -- closing the loop from "found in testing" to "fixed in production"
- A/B testing of prompts, fallback phrasing, and nudge confidence thresholds against real traffic before committing to one configuration
- Native-speaker and compliance/legal review of every Filipino/Bahasa Indonesia string and every compliance-guardrail claim before this reaches a real customer

10. Closing

Everything described here is real, tested code -- 222 tests across four components, all passing, all runnable with `make test`. Every number in this walkthrough came from a report file under docs/, not eyeballed. Where something isn't real -- no live Vapi, no live ASR provider for Q3, illustrative-only numbers -- that is stated in the same breath as the result, because that was the standard this build was held to throughout.

Full detail on every decision: docs/DECISIONS.md. Plain-language defense of the biggest choices plus 15 likely interview questions: docs/DEFENSE_NOTES.md. Line-by-line audit against the assessment brief, including what's genuinely not done yet: docs/SUBMISSION_CHECKLIST.md.